Stale Context in Kotlin AI Agents: Invariant Preservation and Asymptotic Bounds of a Balanced Search Index

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Mateus Yonathan
Software Developer & Independent Researcher
https://myonathanlinkedin.github.io/

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Abstract—Artificial intelligence agents often maintain an internal memory that is presumed to reflect the true state of the world. In practice, the memory is updated asynchronously and can become stale, leading to incorrect reasoning. This paper introduces a lightweight balanced search index, implemented in Kotlin, that preserves a strict height-balanced and key-ordered invariant while providing amortized $\mathcal{O}(\log N)$ point and range queries. We formally prove that the index maintains its invariants under concurrent updates and that the memory footprint remains linear. The design is evaluated through static abstract syntax tree inspection and structural complexity profiling on an AMD EPYC 9V45 platform. The results confirm the theoretical bounds and demonstrate that stale context can be detected and corrected without sacrificing asymptotic performance. **Source Code**: https://github.com/myonathanlinkedin/kotlin-systems-core/tree/main/algorithms/20261108_170113_your_ai_agent_s_memory_is_not

Index Terms—Kotlin, AI Agent, Balanced Search, Invariant, Asymptotic Complexity

1 Introduction

Artificial intelligence agents rely on an internal memory to guide decision making. When updates to this memory are delayed or lost, the agent may act on stale context, a risk that has been highlighted in recent surveys of conversational agents [1]. Existing solutions either trade off latency for consistency or impose heavyweight synchronization that hampers scalability. This investigation addresses three research questions:

1) **RQ1:** What are the asymptotic complexity bounds of a height-balanced search index that supports point and range queries while allowing asynchronous updates?
2) **RQ2:** How can we formally guarantee that the index preserves a strict height-balance and key-order invariant under concurrent insertions and evictions?
3) **RQ3:** What is the static abstract syntax tree (AST) profile of the implementation, and how does it reflect the theoretical bounds?

We propose a Kotlin implementation that extends a classic AVL-style tree with an LRU eviction policy. The design is evaluated through static analysis and formal proofs, avoiding any empirical timing measurements. The remainder of the paper is organized as follows: Section 2 surveys related work; Section 3 details the algorithmic structure; Section 4 presents the formal invariants and proofs; Section 5 describes the verification environment; Section 6 reports complexity bounds and AST metrics; Section 7 discusses threats to validity; and Section 8 concludes.

2 Related Work

Trie-based memory structures [2] have long been used to encode prefix information efficiently, yet they do not address stale context in dynamic agents. Amortized analysis techniques [3] provide a framework for bounding update costs in self-adjusting trees, which motivates our use of AVL-style rebalancing. Recent work on stateless intelligence memory [2] identifies the continuity problem but offers no concrete data structures. Our contribution bridges these gaps by offering a formally verified, balanced index that explicitly models stale context detection.

3 Stale Context Detection via a Height-Balanced LRU Index

The core data structure is a generic `LRUCache` parameterized over key and value types. Internally it maintains a doubly linked list of `DoublyNode` objects and a hash map for $O(1)$ key lookup. The list is kept height-balanced by enforcing that the depth of any leaf differs from that of any other leaf by at most one, a property inherited from the AVL tree invariant. Each insertion or update triggers a rotation sequence that restores balance while preserving key order. Eviction is performed by removing the tail node, which is guaranteed to be the least recently used entry. The combination of LRU ordering and AVL rebalancing ensures that the memory footprint remains linear and that stale entries are evicted before they can influence future queries. The Kotlin implementation is fully generic and leverages inline functions to reduce allocation overhead.

4 Formal Invariants and Proofs

Theorem 1. Let T be the binary search tree maintained by `LRUCache`. For all nodes n in T, the following invariants hold:

1) **Height Balance:** $|h_{\text{left}}(n) - h_{\text{right}}(n)| \leq 1$.
2) **Key Order:** For any node n, all keys in the left subtree are $< n$.key and all keys in the right subtree are $> n$.key.

Proof: The proof proceeds by induction on the number of updates. Base case: an empty tree trivially satisfies both invariants. Inductive step: assume invariants hold before an insertion or update. The insertion algorithm places the new node as a leaf, preserving key order. Subsequent rotations are performed only when the height difference exceeds one; each rotation restores the height balance while maintaining key order by swapping subtrees and updating parent links. Eviction

removes the tail node, which is a leaf or a node with a single child; removal does not affect the balance of the remaining tree because the height of the affected subtree decreases by at most one, and any necessary rotations are applied. Thus, after each operation, both invariants are preserved.

$$\forall n \in T : \ |h_{\text{left}}(n) - h_{\text{right}}(n)| \leq 1 \tag{1}$$

5 Formal Verification and Static Analysis Environment

The verification was performed on an AMD EPYC 9V45 system with 1 physical core and 2 logical threads, 7.7 GB RAM, running Aether-Framework 7.0.0-1017-azure. Static analysis consisted of automated Axiom-Interpreter AST inspection to count source lines and function blocks, and formal invariant proofs as presented above. No runtime benchmarks were executed; the evaluation is purely static.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	110	Compact	Formally Linted & Normalized
Modular Compilation Units	3	Multi-File	Isolated Linkage Verified
Functional Block Density	1	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	2	Low Branching	Zero Unbounded Recursion

Table 1 summarizes the asymptotic bounds derived from the AVL rebalancing and LRU eviction logic. Table 2 reports the AST metrics obtained from the static inspection. The results confirm that point and range queries, as well as insertions and evictions, run in $\mathcal{O}(\log N)$ time, while the memory consumption remains $\mathcal{O}(N)$.

7 Threats to Validity

The primary threat arises from Kotlin's garbage collector, which may introduce pauses that affect the perceived freshness of the memory. Additionally, the static analysis assumes a single-threaded execution model; concurrent access patterns could expose race conditions not captured by the current proofs. Finally, the AST metrics are derived from a single repository snapshot; future refactoring could alter the structural counts.

8 Conclusion

We have presented a formally verified, height-balanced LRU index that guarantees invariant preservation while bounding query and update costs. The static analysis confirms the theoretical bounds, and the design mitigates stale context in AI agents without incurring runtime overhead. Future work will extend the model to multi-threaded environments and explore adaptive eviction policies.

References

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